

Synergy between aggregated anonymized real-world data and agent-based modeling: A case study of the collapse of the Carola Bridge in Dresden

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Extended Abstract¹

In the early morning of 11 September 2024, the Carola Bridge, an arterial road (Bundesstraße) in the city centre of Dresden, collapsed (see Figure 1). The disruption on this major transport infrastructure had a significant impact on traffic flows in the city, and several public transport lines were also forced to be rerouted. Several studies have investigated the impacts of this major disruption (Ijaradar et al., 2026; Lißner et al., 2026). These studies, together with the datasets they make available, serve as valuable resources for validating and further fine-tuning the state-of-the-art MATSim Open Dresden Scenario (hereafter referred to as the Dresden Scenario) (Meinhardt et al., 2026), which has been developed using a recent modelling approach, with the plans of agents (i.e., population) derived from mobile phone data (Neumann & Balmer, 2020). In turn, the agent-based model can provide additional insights at the individual level, contributing to a more comprehensive understanding of behavioral changes that cannot be directly observed from the available empirical data.

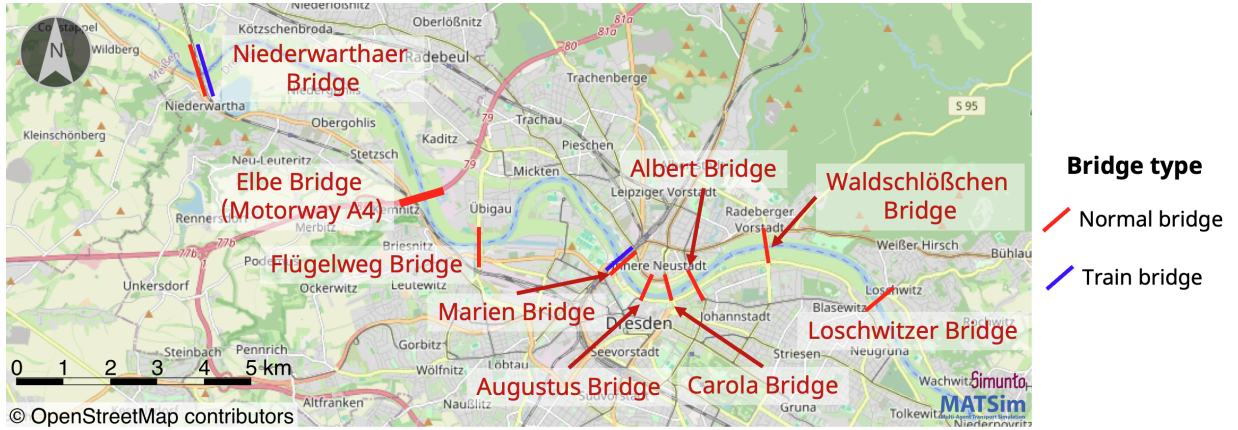


Figure 1: The bridges in Dresden

Within the framework of the VAMOS project, we obtained traffic count data for the bridges before and after the collapse of the Carola Bridge (Landeshauptstadt Dresden, 2026). The local public transport operator, Dresdner Verkehrsbetriebe (DVB), also provided the public transport (PT) network and route information for the periods before and after the bridge collapse. Furthermore, DVB provided estimates of vehicle occupancy for bridge-crossing vehicles, derived from various sensor data, enabling a comparison between the simulated and observed passenger flows across the bridges.

In this study, we incorporate the newly available data described above into the state-of-the-art Dresden Scenario to further refine its accuracy and representativeness. The modifications made to the scenario are summarized below:

¹This extended abstract is submitted to the MATSim User Meeting. Therefore, a general introduction to MATSim is omitted. Readers may refer to Horni et al. (2016) for more details.

- **Calibration of bridge traffic counts:** Bridge traffic count data were used to further calibrate the traffic distribution across the network. By incorporating additional knowledge about the bridge infrastructure, we refined the capacity settings of several bridges, resulting in a better match between simulated and observed traffic counts. This provides a stronger basis for the analysis on travel behaviour and route choice changes following the collapse of the Carola Bridge. Specifically, we reduced the capacity of the Albert Bridge, Carola Bridge, and Marien Bridge, whose effective capacities are lower than those directly inferred from OpenStreetMap (OpenStreetMap contributors, 2026). We also increased the free-flow speed and capacity of the Waldschlößchen Bridge to better reflect the higher traffic volumes observed.
- **Modification of the post-collapse PT schedule:** Following the collapse of the Carola Bridge, several PT lines were rerouted. To reflect these changes, the PT schedule was updated based on the actual post-collapse operations. Rather than generating a new transit schedule from the GTFS data for the whole scenario, we modified the existing transit schedule. This approach minimises the influence of unrelated network changes, such as other construction works or temporary service adjustments, allowing the analysis to better focus on the effects of the collapse of the Carola Bridge.
- **Separating bicycle and motorized vehicles congestion:** Being one of the latest MATSim scenarios, bicycles are explicitly simulated on the network in the Dresden Scenario instead of being teleported. This enables a more realistic simulation of bicycle user behaviour. Currently, the `PassingQ` mechanism is used, allowing faster modes (e.g., cars and trucks) to overtake bicycles on network links. In most cases, the current model works very well. On heavily congested roads, however, bicycles are also involved in the congestion, as they are not allowed to move through the queue. As the collapse of the Carola Bridge induces additional traffic on the neighbouring bridges and cause more congestion, the impact on bicycles may be exaggerated under the current model. In this study, we evaluate two approaches to separate bicycles from the congestion of motorized vehicles: using the `SeepageQ` mechanism and creating a dedicated bicycle network. Both approaches successfully separate bicycles from motor vehicle congestion, each with its own strengths.

After incorporating the above-mentioned modifications, we perform simulations to identify behavioral changes of the population resulting from the bridge collapse, some of which are not directly observable in real-world data. Here, we summarize some of the key contributions that the agent-based model brings to this comprehensive analysis:

- **Changes in traffic volumes on unobserved roads:** The traffic count data provided by the city only covers certain roads. Therefore, traffic volumes on many other roads, including one motorway bridge, are not available. The agent-based model fills this gap by providing additional insights into changes in traffic volumes on these roads. Simulation results indicate that only a small number of car users reroute to the unobserved motorway bridge, confirming that the reduced traffic volumes on the observed bridges are not primarily caused by rerouting.
- **Changes in passenger volumes on railway bridges (S-Bahn and regional trains):** The passenger volume estimates provided by DVB only cover its own services and therefore do not include S-Bahn and regional trains. There are two railway bridges within the study area, one of which is located in the city center. The agent-based model indicates that approximately 5,500 additional passengers, corresponding to an increase of 10.5%, travel on trains crossing the railway bridge in the city center after the collapse of the Carola Bridge. This provides additional insight into the behavioral changes of public transport users.
- **Route choice and mode choice behavior of the individual agents:** The agent-based model indicates that around 10% of the car users and 20% of the bike users switch to other modes, with majority of them switchign to PT, after the Carola Bridge collapsed. Most of the remaining car trips that travelled on Carola Bridge before switch to Albert Bridge, Marien Bridge, and Waldschlößchen Bridge. Due to the fact that some bridges in the city center are already reaching

their capacity limit before the collapse of the Carola Bridge, the extra car traffic causes congestion on those bridges during peak hours in the simulation, which can also be observed in the reality to some extent.

- **Impact on different population groups:** The agent-based model suggests that the travel time of the whole Dresden Scenario increases by 0.96% after the Carola Bridge collapsed. For residents in Dresden, this value increases to 1.16%. For the trips previously crossing the Carola Bridge, the average travel time increases by 10.19% after the Carola Bridge collapsed, assuming the same trips are still to be taken. The collapse of the Carola Bridge also has impact on the trips crossing the other bridges, leading to 3.48% of extra travel time on all the bridge-crossing trips in Dresden. And for non-bridge crossing trips, the extra travel time is only 0.47%.

The contributions of this study are not limited to this specific scenario but are also applicable to future research using the agent-based modeling approach. The results demonstrate that calibrating critical network links can have a substantial impact on traffic distribution. The study also introduces a new approach to traffic count calibration that balances the level of detail and computational workload. By incorporating real-world observations and local knowledge of critical road segments (e.g., major corridors and bridges) into the MATSim network, a more refined traffic distribution model can be achieved, providing a more solid foundation for policy studies. At the same time, this study illustrates the value of conducting a detailed analysis of behavioral changes. With a better understanding of the potential impacts on different individuals and population groups, transport systems can be designed to be more robust, maintaining a high level of accessibility and equity even under unexpected disruptions.

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